

Personalized Learning System PRD

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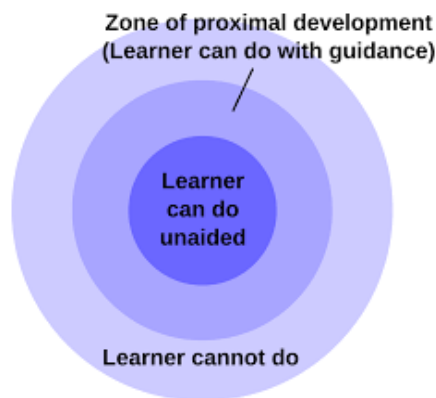
Overview

This document serves as an updated product requirements document for the PLS after the design of the prototype. It explains the goals, the design and the principles of learning science and pedagogy upon which the PLS is based.

The problem of personalized learning

What is personalized learning? We mean the same kind of experience for the learner that occurs with a one-on-one tutor who is using mastery learning-based pedagogy. [Mastery learning](#) dictates that we frequently assess the mastery level of students, and only allow the student to progress to new learning topics once they have demonstrated mastery. This method critically ensures that the learner stays within the “[zone of proximal development](#)” as they learn.

Personalized learning is somewhat of a holy grail in education. We have known since the work of Bloom in the 1980's that one-on-one tutoring using the methods of mastery learning can produce [two-sigma gains in learning](#) for students. The problem of personalized learning is how to reproduce the gains of one-on-one tutoring with mastery learning at scale, for large groups of students. The PLS is a learning tool designed to support personalized learning at scale.



The zone of proximal development (wikipedia)

To achieve personalized learning for a student there are several requirements. We must 1) know the student's learning goals at multiple time scales (near, medium and long term) 2) know the student's current state of mastery of those learning goals, and be able to track changes to mastery over time, 3) have curriculum, learning materials and assessment available for those goals that fall within the zone of proximal development for the student at their current state of mastery, 4) have a teacher available who possesses the skills and knowledge for those learning goals, is aware of the state of mastery of the student, has access to those appropriately-leveled curriculum and learning materials, and is able to transmit the relevant skills and knowledge to the student using those materials as an aid while staying within the zone of proximal development for that student. The PLS is designed to fulfill these requirements for personalized learning for an individual student.

One challenge with providing personalized learning is the lack of available learning materials that are within the zone of proximal development (ZPD) for a learner. With content authored solely by humans it has not been possible to have learning content and assessment available to meet every individual learner's needs, tailored to their ZPD. The advent of generative AI allows us to solve this scaling bottleneck. The ability of generative AI models to both transform existing content as well as generate new content on demand enables the efficient production of learning experiences that fall within the ZPD for that learner. The PLS uses generative AI to produce learning content on demand that is informed by the current mastery level of the student.



Another challenge is the lack of available teachers that have both the requisite domain knowledge and skills for the learning goals of the student, as well as knowledge of the student themselves. AI tutors based on large language models (LLMs) can supplement human teachers and help to overcome the scaling bottleneck of the lack of teachers, by providing instruction and guidance that respects the current state of mastery of the student. But in addition to a student's state of mastery, an effective teacher must know much more about the student, including the student's learning constraints, such as time and budget, and their state of affect – how they feel about what they are tasked with learning. Do they have anxiety, confidence, enthusiasm or lack thereof about individual skills or topics or the subject in general?

The PLS is designed to provide a rich data and input context for LLM-based AI tutors, providing a comprehensive picture of the student, and direct access to learning materials generated from source material provided by a human instructor. With models now able to process multimodal input in real time, we can also provide visual and auditory input to help them perceive the learner's affect as a human tutor would.

Personalized learning does not solve all the problems of education. In particular the problems of “what should students learn” and “how do we help learners set learning goals” are not directly addressed by personalized learning and the PLS. Later we will discuss those problems and how the PLS can be used within a broader education system that addresses those critical areas. But first we explain the goals and design of the PLS.

Goals

These are the goals of the PLS in three broad categories.

A) Support personalized learning for learners

According to our formulation above, these are the capabilities that are needed to support personalized learning for a student:

1. Support mastery learning for students
 - 1.1. Store and display the skills and knowledge necessary for a learner learning goals
 - 1.1.1. Provide visibility to the student of the knowledge and skill graphs associated with the near term, medium and long term learning goals for that student.
 - 1.1.2. A student enrolled in a course should be able to see the KG and SG for the immediate time frame (e.g. current week) of the course, the entire course, and the program they are enrolled in.
 - 1.2. Track mastery for the learner across the knowledge and skills and reflect it back to the learner
 - 1.2.1. Use events from learner interactions to model mastery for students over time
 - 1.2.2. Make mastery progress over time visible to the student across the skill and knowledge trees, to inform and motivate them and help identify challenge areas
 - 1.3. Provide feedback to the learner on the level of effort and time required to achieve mastery
 - 1.4. Recommend learning goals to the student based on their current mastery and salient topics
 - 1.4.1. These recommendations effectively mimic a learning coach for the student or an [“outer-loop tutor”](#)
2. Given a learner and a learning goal, produce a personalized lesson for that learning goal that falls within the current zone of proximal development for that learner
 - 2.1. The lesson should include a set of learning materials in various modalities (video, text, audio) depending on the preference of the student
 - 2.2. The lesson must include formative assessment: opportunities for the student to practice the skills and assess their knowledge for the learning goal, and for the system to acquire evidence of student mastery
3. Support the student with instruction, guidance and feedback as they proceed with that personalized lesson

B) Help educators scale teaching

The PLS is a tool for educators as well as students. Consistent with the [AI principles of Google](#), we believe that a learning system using AI must be accountable to people. In education this means that human educators must be in control of the learning process, including defining the skills to be learned, curating relevant learning content used as source material for generating lessons for students, and reviewing and curating generated assessment items. Humans set the priorities for learning, and make all important decisions.

We recognize the unique power of human educators to instruct and advise students. We would like to reduce the burden and enhance the efficiency of educators in their daily tasks, allowing them to reach more students directly.

These capabilities of PLS are associated with the goal of helping educators scale:

4. Generate curriculum and learning materials for teachers
 - 4.1. Materials include lesson plans, assessment, and learning content. Automatic generation of learning content is one of the key values of AI in education. This capability alone can save educators significant time and effort, allowing them to spend more time on instruction and coaching. We believe educators should review any assessment that is generated to ensure quality and alignment.
 - 4.1.1. The same capabilities used to generate personalized learning plans for students can be used to generate generic learning plans for cohorts of students.
 - 4.2. Offer better tools for curriculum design that leverage skill and knowledge graphs
 - 4.2.1. Allow educators to define groups of skills and knowledge as modules, courses, credentials and programs
 - 4.2.2. Enable scalable creation of summative assessment for those groups
 - 4.2.3. Enable dynamic creation of learning content for those groups
5. Allow educators to curate the source learning materials used by students
 - 5.1. We have heard repeatedly from educators that this is important. There are many sources for learning any particular topic, and many variations in presentation. Educators want to curate the content used as ground reference by students in a course they manage.
6. Provide insight to educators (instructors, counselors, curriculum designers) as to the mastery progress of students using the system.

C) Help LLM tutors be effective

The PLS is designed to be a platform and surface for LLM-based AI tutors, providing them a rich data context to help them be more effective in guiding and instructing students.

7. Provide data context for LLMs including
 - 7.1. student mastery and affect, registration history, academic history
 - 7.2. skill and knowledge graphs for the learning context
 - 7.3. learning content (textbook, video, assessment)
 - 7.4. learning context (course schedule, degree program)
8. Allow for multiple AI personae with different contexts
 - 8.1. Academic coach
 - 8.1.1. answer the question “what should I study today”
 - 8.1.2. teach metacognitive skills and encourage students to practice
 - 8.2. Skill and knowledge tutor
 - 8.2.1. providing feedback to student assessment attempts
 - 8.3. Interview simulator

- 8.3.1. This is an example of an LLM agent that can act as a learning activity for students. Many more are possible.

D) Provide learners access to GenAI tools that is equitable and easy to use

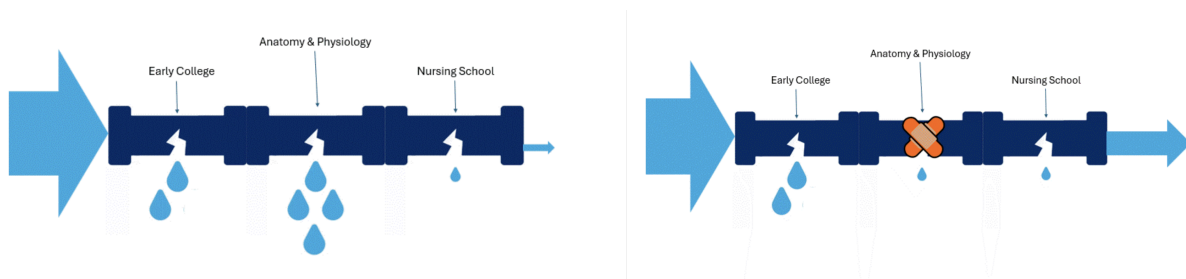
Inequitable access to GenAI is a little recognized problem. Because many of the advanced features of GenAI require a paid subscription, not all students may have access to these advanced features. A school system should provide equitable access for all students to these tools. PLS provides access to foundation models like Gemini that ensures all students have the same access.

The PLS can act as a “workbench” application for GenAI applied to education, similar in spirit to NotebookLM, or an IDE for developers, but tailored for education. One challenge for students is that they don’t have unfettered access to the electronic versions of learning materials like their textbook, and so loading such content into GenAI is often not possible. In the PLS the content is always provided as context to the GenAI models.

9. Provide equitable access to GenAI for students
 - 9.1. PLS allows for school system-wide equity in access
10. Provide a workbench for GenAI for education
 - 10.1. Make it easy for students to load learning content into models
 - 10.2. Store frequently used prompts and workflows for a student

Metrics and benefits

The primary metrics used by stakeholders to evaluate the PLS will be course completion, final grades, retention and program completion (graduation rates). Incremental improvements in these metrics of just a few percent can have a large impact on downstream metrics, like the number of qualified learners available for job placement or further studies. As an example for the University System of Georgia, an analysis was done on improvements in the passing rates of Anatomy and Physiology, a bottleneck course that has passing rates in many institutions below 50%, and yet must be completed to progress in a healthcare related program. An improvement in the passing rate of 5% across the system resulted in 600 additional students available for nursing school annually. There is a chronic shortage of nurses and a corresponding shortage of qualified candidates for nursing school. This benefit alone was seen as sufficient to justify piloting the PLS.



A diagram created by the USG team to show the benefit of improvement in pass rate for Anatomy and Physiology

It is worth noting that the “local impact”, meaning the impact on the proximal success of any learner, provided by any learning tool like the PLS is likely to be incremental - a few percentage points improvement. This is because most of the reasons that students fail to succeed are beyond the ability of a learning tool like PLS to influence. Structural problems in the learning system have a much larger impact on student success, like the lack of control of pace (e.g. students must learn according to the schedule of a semester calendar) or the lack of visibility into preparedness (students enroll in courses that they are not prepared for from a mastery standpoint, so they cannot enter the zone of proximal development). See the section below on “why students fail” for a more in depth discussion.

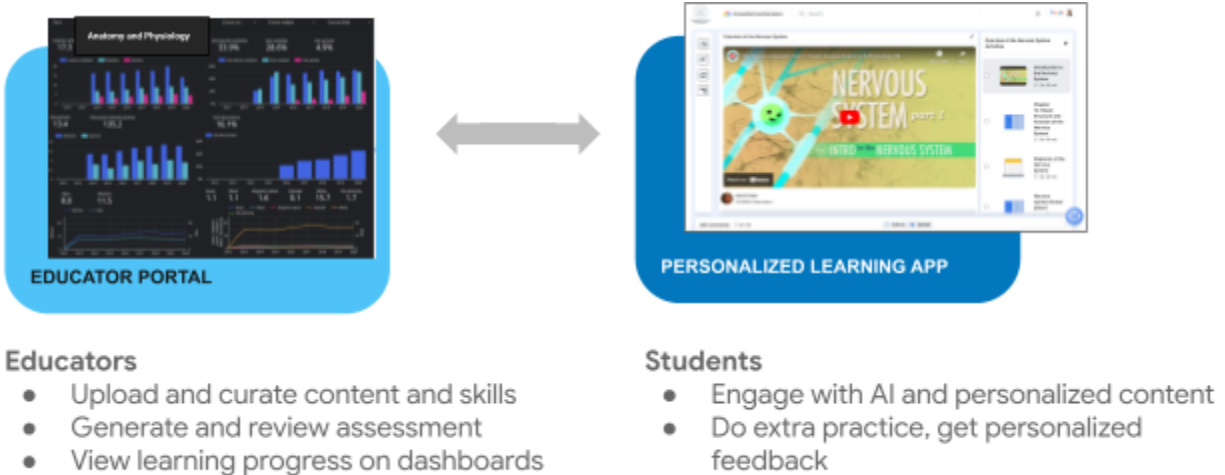
However small percentage improvements can compound. If a tool like the PLS is widely deployed across an education system, like the public university system of a state, the impact on graduation rates and learners' preparedness for careers can be significant.

Additional benefits related to curriculum management have been noted by the USG team. For a course offered across a university system, the exercise of loading content and generating knowledge graphs, defining syllabi and skills and a course schedule, will have tremendous value for the system by centralizing and standardizing the curriculum for that course. There is no such centralized representation of curriculum across the state now. Every institution manages their own curriculum for a course like Anatomy and Physiology, which is taught at 22 out of the 26 USG public universities. The PLS will allow a centralization and standardization for curriculum, which also greatly expands the possibility and impact of learning analytics.

Optimizing *what* students learn is actually more important for the metrics society cares about, such as the time and cost of reskilling for a new job, or the availability of workers for key industries like healthcare and cybersecurity. But that optimization is beyond the purview of the PLS. See the background section for a discussion.

The PLS solution

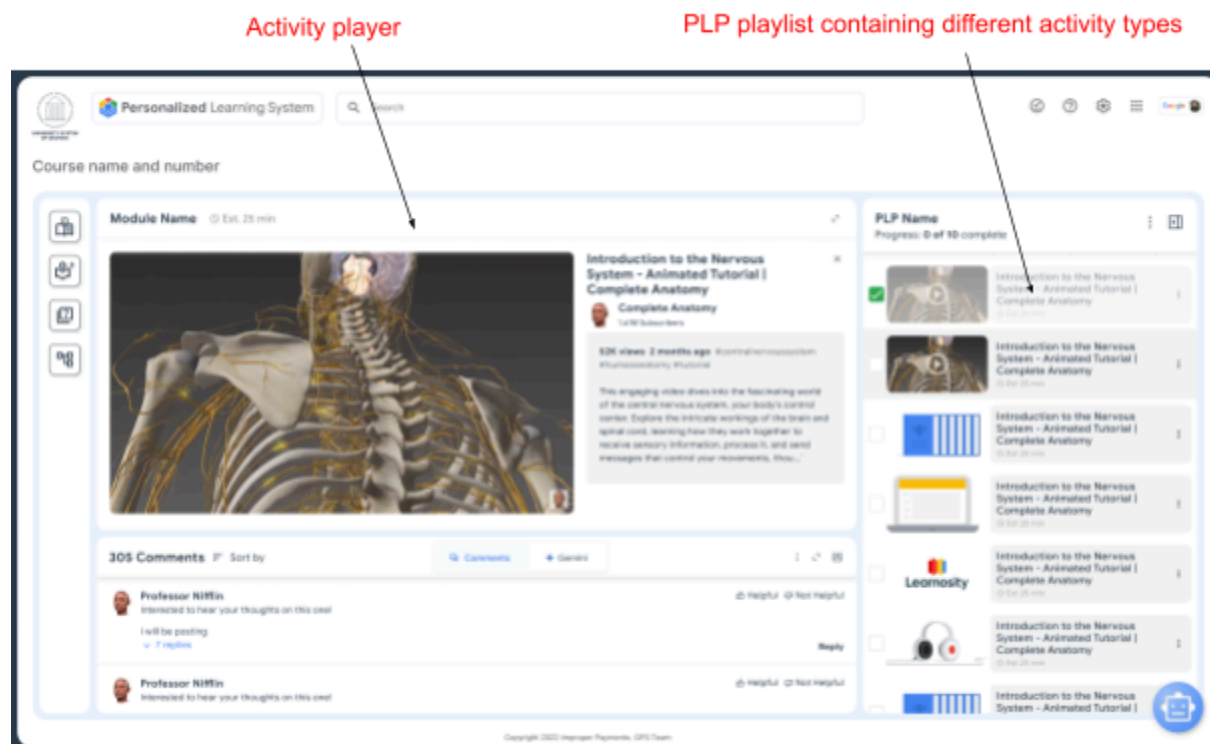
The Personalized Learning System (PLS) consists of two web applications. There is a learning application, and an educator portal. We focus first on the learning application.



The apps of the personalized learning system

Personalized Learning App

The learning app of the PLS allows students to request and complete “personalized learning plans” (PLPs), which are lessons designed to help a student learn a particular topic. The PLS allows the student to create PLPs for any topic on demand. The PLPs can be customized by the student to accommodate their learning preferences, including desired time, difficulty and reading level, level of assistance from the AI instructor, as well as modality of presentation. These parameters help to ensure that the PLP is within the current zone of proximal development for the learner. The parameters may also be chosen automatically by the PLS, based on the mastery profile for the student.



The main screen of the PLS prototype. The current “PLP” is displayed on the right side, as a “playlist.”

The PLPs are drawn from learning content (e.g. textbooks, lecture videos or other instructional videos, and assessment) that are curated by an instructor or curriculum manager. This happens via the “educator portal”, about which we go into more detail below.

The PLS learning app is explicitly designed to mimic the user experience of YouTube. This user experience is one that all users will be familiar with, and indeed are likely already using as a learning tool. In fact, in a basic sense this app is an attempt to turn the YouTube UX into a generic learning app, incorporating AI in a seamless way, and allowing the user or an instructor to create personalized playlists, or PLPs, on demand.

Personalized Learning Plans

A PLP is a personalized lesson for the learner, created either by the learner or by an instructor. The student may have multiple PLPs active at a given time. The student uses the PLS by creating and completing PLPs.

A PLP consists of an ordered list of “learning activities.” A learning activity consists of learning content of different media types. Each learning activity is delivered to the student in the PLS via an “activity player” which supports the media types associated with that activity. For example, for video media the player can be the [YouTube Player for Learning](#). The student completes the PLP by viewing, reading, listening and interacting with each learning activity in the list, via the respective learning content player for that activity.

The diagram below depicts the process for generating a PLP.¹



The process of generating a PLP

Let's dive into each step in more detail.

1) Choose learning topic and learning parameters

The PLS is designed to support the learning journey of the student by recommending to them the most important concepts or skills they should study next. In the current design these recommendations are provided in the menu used to create a new PLP (see below). These recommendations take the form of individual skills or concepts, or learning goals (which are just groups of skills and concepts) drawn from the learning goals of the student. How are these recommendations made? The PLS has a representation of the skill and knowledge graphs of the course², and based on those graphs as well as the mastery model for the student, can know what the most salient nodes are for the student at a given point in time.

In addition to representing the mastery for the student across all nodes in the skill and knowledge graphs, the mastery model stores the accumulated evidence for that mastery. The evidence takes the form of student attempts at assessment items labeled with the respective skill or knowledge concept. This allows the mastery model to represent a confidence level for each mastery value. This suggests a simple algorithm for making recommendations for PLP topics:

¹ This process bears many similarities to a multi-modal RAG pipeline. In some sense a PLP can be thought of as the output of a multi-modal RAG query that includes generation parameters to control the leveling of the content to ensure it falls within the zone of proximal learning for the learner.

² See the background section for more detail on skill and knowledge graphs.

*retrieve skill and knowledge nodes for active learning
goals for student*



rank nodes by salience



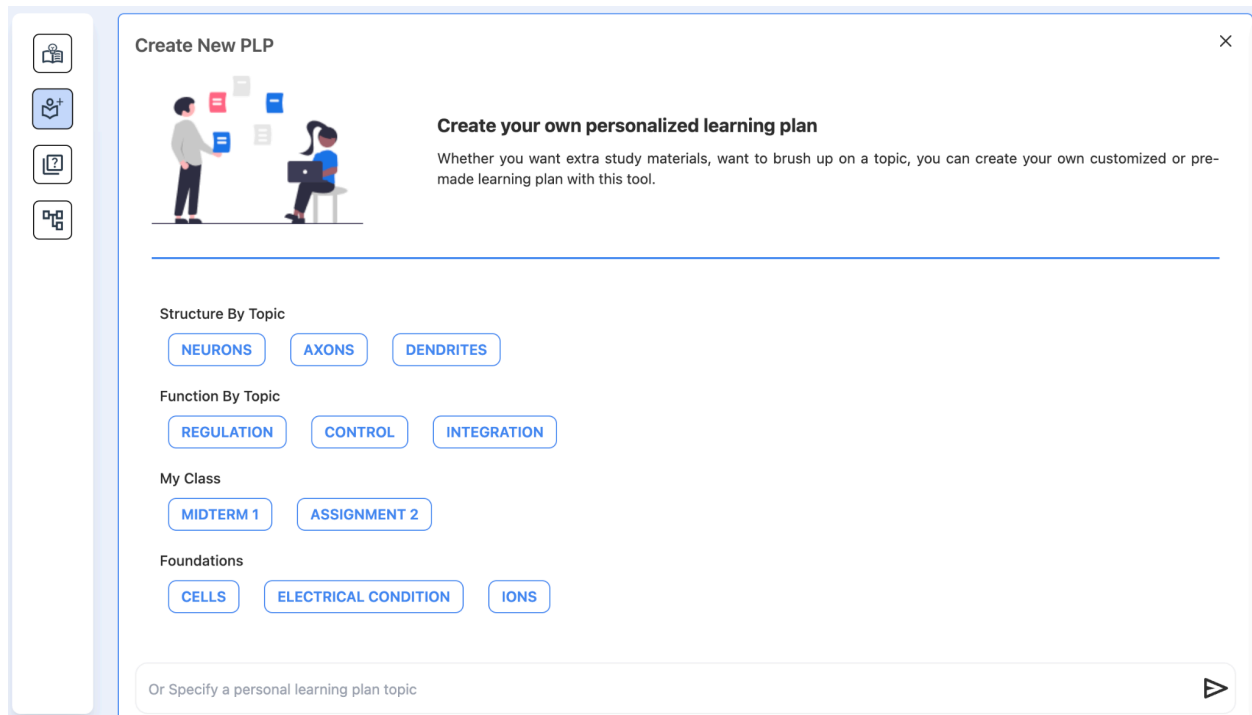
*weigh rankings by
 $1/\text{mastery} \times 1/\text{confidence}$*



take top K nodes

A simple algorithm for recommending PLP topics to a learner

These topics are surfaced to the user when they open the PLP creation menu.



Create New PLP

Create your own personalized learning plan

Whether you want extra study materials, want to brush up on a topic, you can create your own customized or pre-made learning plan with this tool.

Structure By Topic

NEURONS AXONS DENDRITES

Function By Topic

REGULATION CONTROL INTEGRATION

My Class

MIDTERM 1 ASSIGNMENT 2

Foundations

CELLS ELECTRICAL CONDITION IONS

Or Specify a personal learning plan topic

The PLP creation menu in the prototype

In addition to the learning goal, the PLP request must specify the learning parameters for the PLP. These parameters ensure that the PLP stays within the zone of proximal development for the learner. The learning parameters supported by the prototype design include:

- a) difficulty level
- b) reading level
- c) level of AI assistance
- d) amount of time needed to complete the PLP

2) Translate goal into set of knowledge and skills

If the user selects one of the recommended topics, they have by definition specified the knowledge and skills associated with the PLP, since each recommended topic corresponds to a learning goal, which in turn maps to a set of skill and knowledge nodes from the master skill and knowledge graphs for the learning context.

The user also has the option to request a PLP with a natural language request. In this case the system must translate that request into a set of knowledge and skill nodes. This implies a generic capability to parse a natural language request into skills and knowledge. Based on initial experiments we believe that LLMs provided with the appropriate context (e.g. the complete set of skills and knowledge in the learning graph for the course) can perform this parsing effectively.

3) Retrieve and rank relevant learning content

The content ingestion process extracts knowledge from the content and builds a KG for each content item, where the nodes of the KG are linked back to the content to form an index. Skills come from non-content sources (see the background section) but are also indexed to the content during the course setup process. This graph-based indexing is similar to the kind of indexing performed for RAG search pipelines, with the main difference being the nodes of the graph acting as index keys rather than embedding vectors drawn from arbitrary chunks of content.

We will also perform such chunk-based semantic indexing on the content, forming a multimodal RAG index for the content. Given a set of knowledge and skills nodes, and possibly a natural language request from the user, these combined indexes (KG, SG, and semantic) are used to retrieve the relative content from the content store.

Once retrieved the content chunks are reranked. Additional filtering can be applied such as time length limits. The top k form the list of activities for the new PLP.

Assessment items may be directly retrieved according to the learning parameters from the assessment item bank, a database of pre-generated items indexed by skills and knowledge as well as learning parameters of difficulty level, reading level and amount of time needed to complete.

4) Transform learning content

Once content is retrieved and ranked it may need to be transformed according to the learning parameters of the PLP request. This transformation process will be different according to the media type of the content.

text: The most straightforward is text content, which can be transformed by LLMs to different reading and difficulty levels.

video and audio: these can be regenerated from the source content by multimodal LLMs. There may need to be specialized logic for this generation process according to the type of video. For example, for lecture videos, it will be possible to change the talk track of the speaker to alter the difficulty level while preserving the likeness and mannerisms of the original video.

images: given the power of image generation models, it is possible to transform images into simplified or more complex versions. One way to do this would be to extract a comprehensive text description, transform the text description itself according to the learning parameters, then regenerate the image based on the transformed description.

assessment: This transformation process does not apply to assessment which we generally assume to be pregenerated (and reviewed and curated) by human instructors.

Learning activities

These are the basic learning activities supported in the prototype along with the respective players.

Media type	Player
PDF viewer	React PDF viewer
Audio	native HTML audio
Slide viewer	custom PLS slide component
Assessment	custom PLS assessment player

For assessment, we can use an external API service that can create and serve assessment items, like [Learnosity](#).

The PLS is expressly designed to be extensible, to accommodate new activity types. One could add activities for learning coding in Javascript (for example using the P3.js player), or practicing how to interview a stakeholder by interacting with an AI avatar, or designing AI agents.

Learning goals

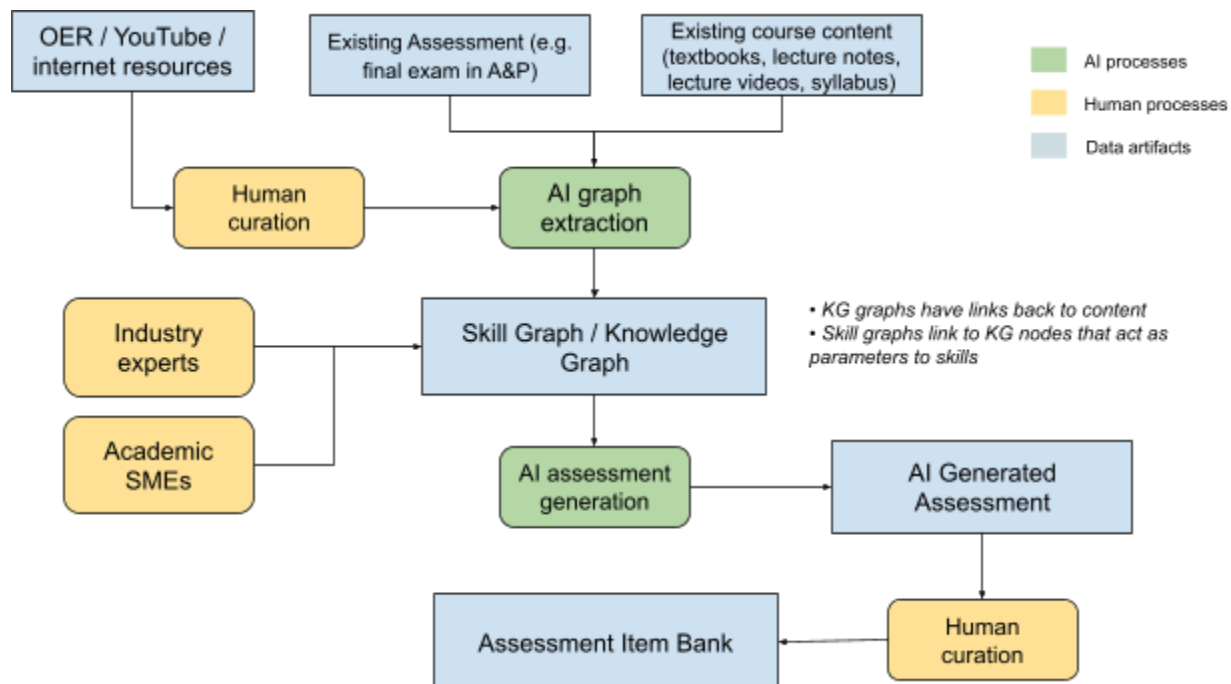
We define a learning goal for a student as a set of skills and knowledge nodes drawn from a KG and SG.

In a time-bound learning environment such as a university course, the KG and SG nodes may be associated with time divisions like “Week N”. It will be natural to create learning goals around those groups of nodes.

There are usually also useful relevant milestones in a course such as summative assessments (quizzes, midterms and final exams). We can create learning goals for those as well. This requires the instructor or curriculum manager to input the course schedule, with summative assessment dates, and specify the skills that will be assessed at each summative milestone. Instructors will perform these tasks in the educator portal.

Assessment

This diagram from the [TDD](#) for the PLS depicts the process of assessment generation.



The assessment generation process is managed in the educator portal.

There is only one type of assessment (free form answer question) in the prototype; many more can be added using an external provider like [Learnosity](#).

Practice Problems

Mastery

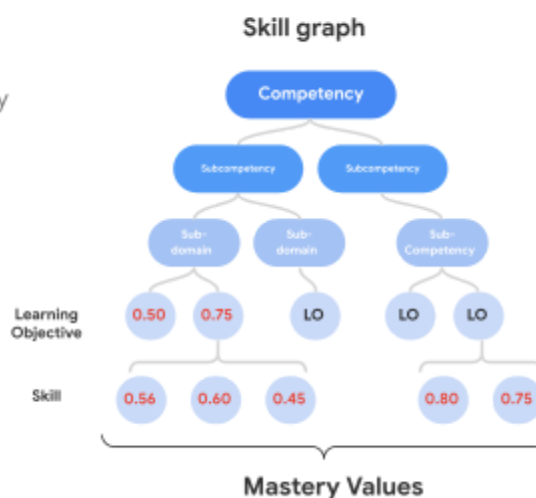
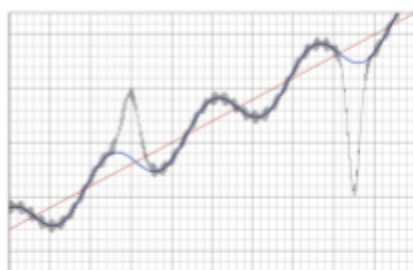
We use the skill and knowledge learning graphs as scaffold data structures to represent the state of mastery for a given student. See the Background section below for more detail on the skill and knowledge graphs. We maintain a mastery model for each student, over the collected skills and knowledge that comprise the learning goals of that student.

Given a set of skill and knowledge nodes that comprise the learning goal of the student, the mastery of the student for each node is defined as the probability that the student will successfully complete an assessment item labeled with that node. Mastery changes over time as the student completes learning activities within the PLS or the other learning experiences in their academic program. This we frame the problem of modeling student mastery as a time series prediction problem, where the mastery of a given node is our prediction that the student will answer the next problem correctly labeled with that node.

Student Mastery can be modeled by two algorithms:

- **IRT** (Item Response Theory)
- **DKT** (Deep Knowledge Tracing)

Mastery for a node is defined as the **probability** the student will answer the next question **labeled with that node** correctly

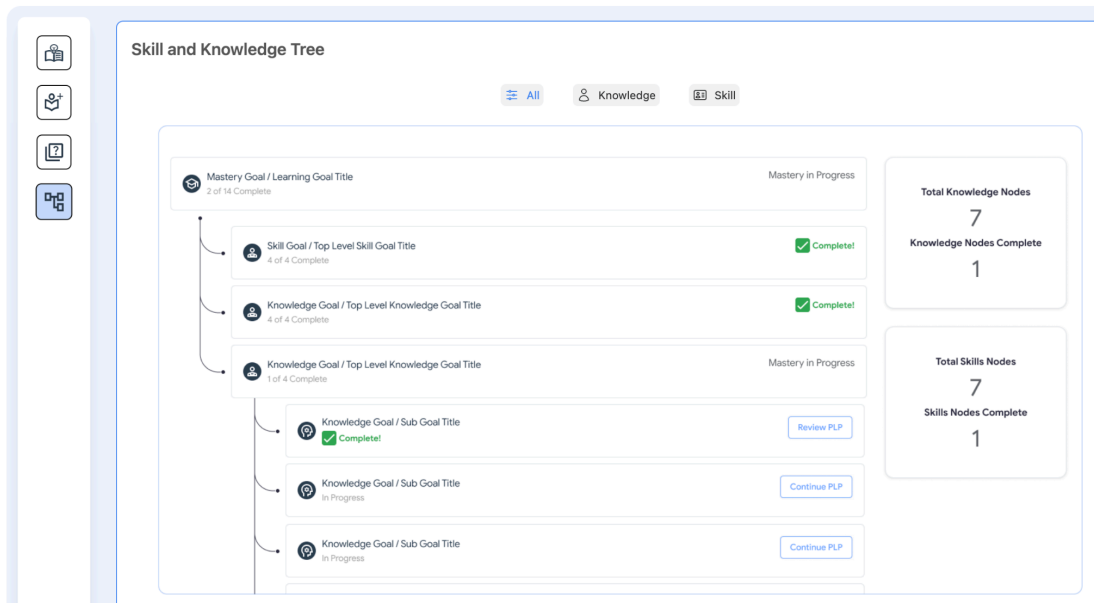


We have previously implemented [Deep Knowledge Tracing](#) in the Cloud Learning Platform, a modeling technique for the mastery time-series prediction that leverages deep neural networks (LSTMs). [Item response theory](#) (IRT), a statistical modeling technique used for decades in education, is also used to impute statistics related to mastery including the relative difficulty of assessment items and their alignment to a skill or knowledge node. The alignment of an item refers to its ability to discriminate between learners who have mastery of a particular skill or concept and those who don't.

Skill and knowledge trees

The skill and knowledge menu in the PLS is intended to provide a view into the skills and knowledge that are relevant to the student at multiple times scales. Mastery can also be overlayed on the skill and knowledge graphs allowing the student to track their progress. Other

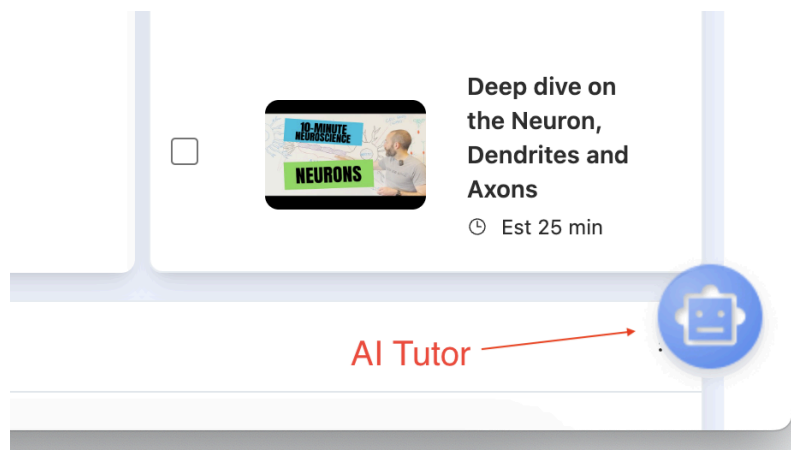
metadata like salience can also be surfaced here. The design of this menu in the prototype is a placeholder: there is still room for design innovation in this part of the interface.



The skill and knowledge menu in the PLS prototype

AI Tutors

One goal of the PLS is to provide a surface and data context for LLM-based tutors. The LLM tutors included in the prototype are meant to serve as examples of how tutors can be integrated into the PLS. After testing with students it is likely that these tutor surfaces will be revised. There may be an integration opportunity for an “ambient tutor” that can bridge the student learning experience from the PLS to other surfaces like the LMS. In some sense the current PLS prototype lacks a real “ambient” tutor, given the tutor chatbot is accessed by clicking the tutor icon.



The AI Tutor icon in the lower left launches the tutor chatbot

There are three places in the prototype where the student can interact with an LLM in a tutoring context:

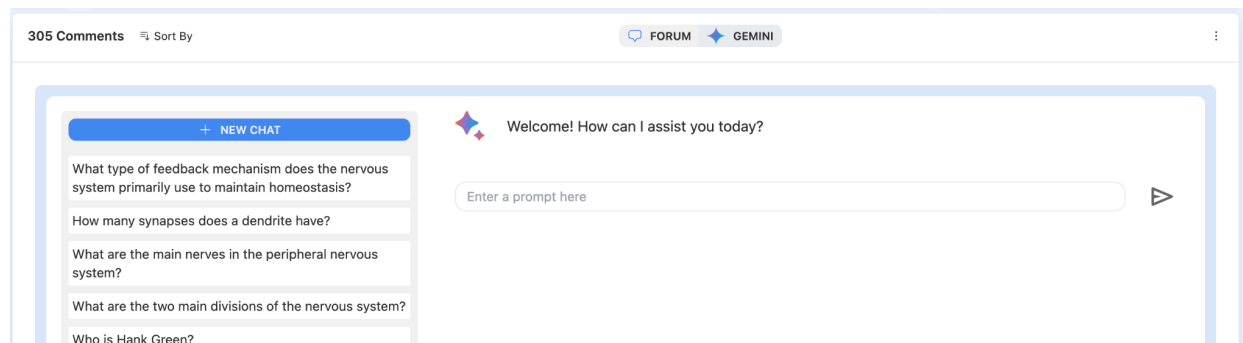
1) Feedback for assessment answers

Scoring and feedback for assessment answers is provided by an LLM. In the prototype this LLM is the “QuizMe” model of LearnLM.

2) Tutor chatbot

There is a tutor chatbot available in the prototype, opened by clicking the icon in the lower left corner. This chatbot will have access to the entire data context for the PLS. This tutor is intended to be an “ambient tutor” in the sense that it can act as a tutor for the entire PLS experience. The chat window interface may not be the ideal UX for such an ambient tutor - the tutor may need to “escape” the confines of the chat interface, so that it could (for example) greet the user when they return or reopen the PLS with advice on “what to focus on next.”

There is also opportunity for a more “personified” tutor here. The same “AI tutor personification” strategy used for the [Shiffbot](#) could be applied to this tutor.



The Gemini panel

3) Gemini panel

The Gemini panel at the bottom of the UX offers a generic interface to Gemini that can allow the user to take advantage of the many [features of Gemini that can support learning](#). This interface can also provide access to these advanced features in a system that is made available to all students, whereas many of the advanced features are only available on the paid tier of Gemini.

The prototype UX stores simple chat history, and does not allow for file upload. Extensions could also store prompts and workflows, to realize the goal of making the PLS app a “GenAI workbench” for education.

Integration into school systems

In a university or K12 environment, the PLS will be accessed via the LMS of the school via LTI or another interoperability standard for external tools (such as Classroom plugins). The PLS will also likely need to integrate with the SIS to access student academic history as well as the full learning context of the student.

Educator portal

The educator portal (EP) is the interface for educators to administer the PLS. There is not currently an implemented prototype for this UX. However instances of this UX were created for earlier versions of the AI tutor based on Cloud Learning Platform, and there are [designs](#) available for the content and assessment management components of this app.

The educator portal will include these capabilities:

1. Manage courses served by the PLS

- 1.1. Course content

The EP allows educators to upload approved content for a course. Content can include textbooks, videos, individual files, slides and images, audio tracks (e.g. podcasts).

- 1.2. Suggest new content to professors

The EP may search for new content in repositories like YouTube or OpenStax and suggest this content to educators.

- 1.3. Course schedules and milestones

Course schedules (e.g. the topics covered in each week) and milestones (summative assessment dates, assignment due dates, etc) provide important context for the PLS and are specified and managed in the EP.

2. Manage skill and knowledge graphs

When content is uploaded the system automatically extracts knowledge from that content. Skills are specified directly by the educator. The educator may upload syllabi that contain learning objectives (a granular kind of skill).

- 2.1. The KG and SG graphs may be browsed and edited in this app.

3. Generate assessment with LearnLM

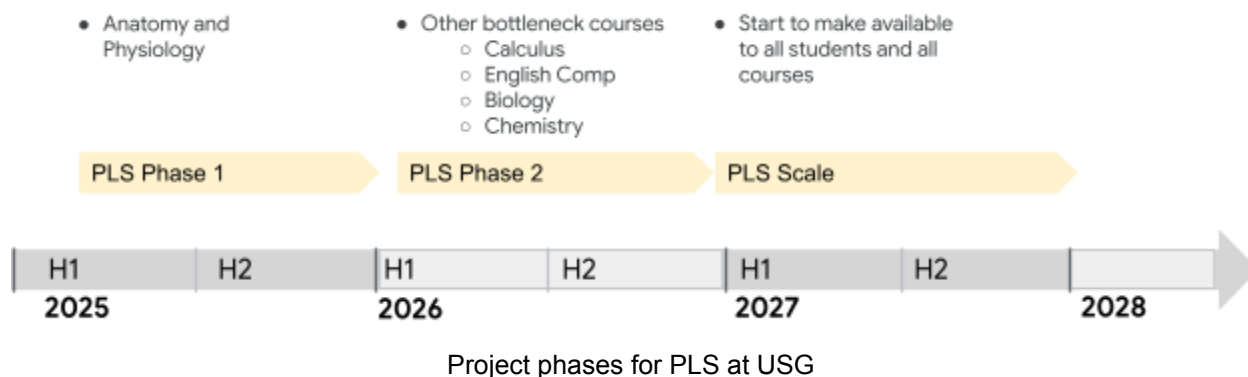
See the [assessment section](#) above for a depiction of the assessment generation process. That process is mediated by a curriculum manager using the EP.

4. Dashboards for student mastery

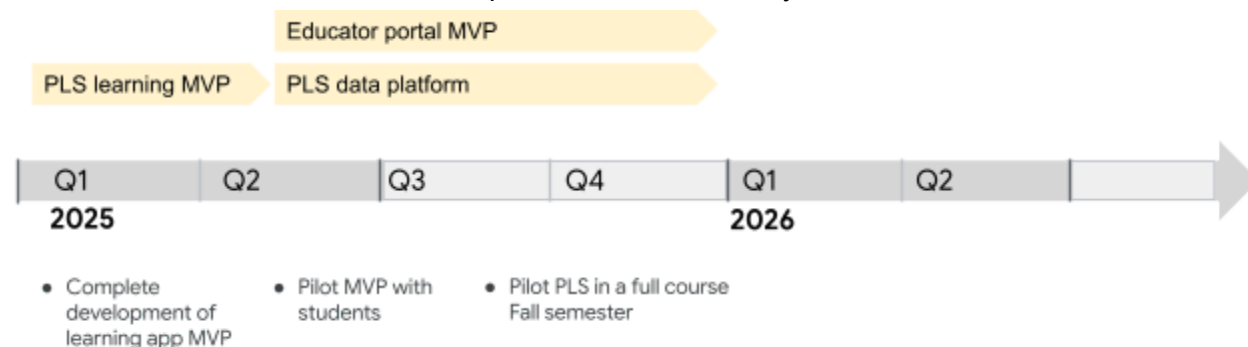
The EP can provide dashboards that reflect mastery across cohorts, for individual skills and concept nodes in the KG and SG.

Roadmap

The project schedule for the University System of Georgia has the timeline below. There will be other customers piloting the PLS in 2025, including Capella at SEI. Several universities have expressed interest in a pilot, including UC Riverside, Northeastern, Rutgers, Galen School of Nursing, Reichman Medical School (Israel).



Our best estimates for feature development for PLS currently looks like this:



Appendix

What should students learn?

There is an additional central problem in education that is not directly addressed by the PLS, namely what should the student be learning?

Students should learn how to learn

Metacognitive skills are not part of the standard curriculum. Yet those skills are the most critical for student success. Students need to learn effective strategies for absorbing large quantities of knowledge, for acquiring new skills, and understanding a new domain. Students need to understand how to prioritize the time they spend on learning.

Students should learn 21st Century skills

There are many skills that are not part of any standard curriculum that are critical for success in any profession. These skills are often referred to as 21st century skills, and include leadership, problem solving and critical thinking, teamwork and collaboration, persuasion, expectation management, and risk assessment.

Students should learn job skills

Job-specific skills are not part of the curriculum, because the education system is designed to prepare students for general careers. Yet we know that many job specific skills are in high demand and low supply.

Why do students fail?

Across higher education, over 35% of students fail to complete the degree programs they enroll in. For many segments the six year graduation rate is less than 50%. Here are some main reasons for student failure. These are offered without attestation, but have been reviewed by education leaders at USG.

- Lack of “academic sophistication”, which is a proxy for inadequate state of preparation and a lack of meta-cognitive skills (“study skills”). “Study skills” include skills to absorb large quantities of facts and concepts, skills to reinforce those concepts and practice applying those concepts, and skills to make study plans and execute them.
- Lack of preparation in prerequisite subjects, leading to students starting a learning experience (a course or program offered at a school) outside the zone of proximal development.
- Lack of visibility into that state of preparation - students don’t know what they need to know, and what they currently know, and educators are similarly in the dark about these aspects of student preparedness and mastery.
- Lack of control of pace, to give students time to remediate their lack of preparation.
- Lack of choice in how students learn, including lack of options in form, format, and modality of instruction and content. In order to stay within the zone of proximal development students need to be able to adjust the the difficulty level of learning content. To make best use of their time and provide a rich learning environment, students need to access learning content in different modalities and formats, such as audio podcast, explanatory video, and simulation environment in addition to the standard “scrollable text with images.”

Background

This section contains some background material for our approach to the problem of personalized learning with the PLS. This includes work that was done previously on [Cloud Learning Platform](#) (CLP) and the first AI Tutor system that was built on CLP.

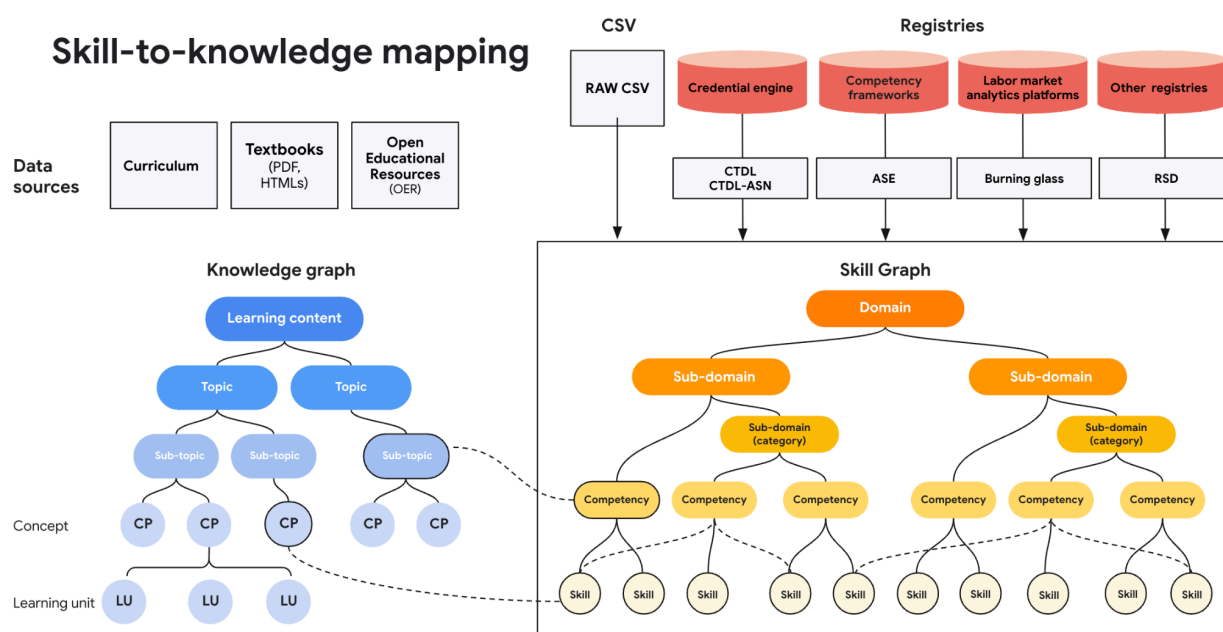
Skills and knowledge

We may broadly define the goal of education as the acquisition of skills and knowledge by a learner. To build a technology system that enables personalized learning, one must have a data representation of the skills and knowledge that a learner can learn using the system. We use the data representations of skills and knowledge to drive the creation of personalized learning

content and learning experiences that fall within the zone of proximal development for the learner, and to measure the time evolution of student mastery.

In the PLS we make a distinction between skills and knowledge, both conceptually and in data structures. [This paper](#) provides the motivation and details of this approach, and how we represent skills in knowledge in data. To summarize briefly, we treat skills as “cognitive programs”, and knowledge as the data that represents the input context and the output for those programs. We represent both skills and knowledge using hierarchical graph data structures, or “learning graphs”, where each node in the graph corresponds to a single skill or knowledge entity, and the children of a node represent a decomposition of the skill or knowledge entity into subskills or sub-entities. The edges in a learning graph represent relations between the nodes in the graph. These relations can be compositional, representing part-whole or parent-child relationships between the nodes. Graph edges can also represent predicate relationships between entities like $A \langle \text{PRED} \rangle B$, for example A (a person) “is the author of” B (a book).

For skill graphs, edges can encode learning precedence relationships like “A must be learned before B.” This precedence often results from a compositional relationship between the skills: you must learn skill A before skill B because A is a “subskill” of B.



Skill and knowledge trees, their data sources, and mapping between them

For knowledge, our learning graphs are similar to the knowledge graphs (KGs) that are widely used in search and information retrieval. However these KGs are organized according to academic disciplines, with the top levels of the graph representing the most abstract concepts of a domain. A KG for a subject like biology will have nodes at the top levels of the graph that represent major subdomains of biology like microbiology, cell biology, ecology, genetics,

anatomy, physiology and so on. Each of those nodes will have children that represent the major topics of that subdomain. There is a natural isomorphism between a KG for an academic domain and the table of contents for an introductory textbook in that domain.

We find it useful to make a mapping between the skill and knowledge graphs, representing the knowledge that may be used as input context for a given skill. This can be used for example to generate assessment activities for skills, by providing multiple examples of how the skill might be applied within a given knowledge domain.

Skills and competencies

Data sources for skills and knowledge

Skills and knowledge come from different data sources